Biological Dataset Selection:
Lung Adenocarcinoma Gene Expression

Project Overview

This project focuses on selecting a publicly available biological dataset for computational analysis of lung adenocarcinoma. The objective is to investigate differences in gene expression between tumor tissue and non-tumor lung tissue using existing genomic data, without conducting laboratory experiments.

Research Question

Which genes show differences in expression between lung adenocarcinoma tumor tissue and non-tumor lung tissue?

Dataset Source

- **Repository:** NCBI Gene Expression Omnibus (GEO)
- **Dataset accession:** GSE10072
- **Dataset link:** https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE10072
- **Organism:** Homo sapiens (human)
- **Data type:** Gene expression profiling using microarrays
- **Technology:** Affymetrix Human Genome U133A Array (GPL96)

Dataset Size

The dataset contains 107 final gene-expression profiles:

- 58 lung tumor tissue samples
- 49 non-tumor lung tissue samples

The dataset includes processed expression measurements for thousands of microarray probes. These are expression profiles from tissue samples, not 107 unique individuals.

Proposed Analysis

The dataset will be explored using NCBI GEO and GEO2R. Tumor and non-tumor samples will be compared to identify genes with differences in expression. The analysis may include statistical significance, adjusted p-values, log fold changes, and visualization of the results.

Expected Outcome

The project aims to identify genes with differing expression levels between the two tissue groups and explore their potential biological relevance to lung adenocarcinoma. The findings will be interpreted in the context of the dataset and its limitations.

Tools and Resources

- NCBI Gene Expression Omnibus (GEO)
- GEO2R
- Spreadsheet software for organizing results, if required

References

1. NCBI Gene Expression Omnibus, GSE10072: https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE10072
2. GEO2R documentation: https://www.ncbi.nlm.nih.gov/geo/info/geo2r.html